TOM 2 20111124 M/S

USN | | | | | | | | | | [22TCSE532]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

ADVANCED MACHINE LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	How to declare a tensors in Tensorflow?	L2	CO1	**02**
	b.	Compare and contrast bagging and boosting.	L2	CO2	**02**
	c.	What is double DQN?	L1	CO3	**02**
	d.	Describe k-fold cross validation	L1	CO4	**02**
	e.	List the type of Serializing method in scikit learn. Which method do you prefer? Why?	L3	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	List type of Activation Function in Tensorflow.	L2	CO1	**05**
		OR			
	b.	What is the purpose of using multiple executors?	L4	CO1	**05**
3.	**a.**	How does the process of resampling with replacement in the bootstrap method work, and what are its implications for bias and variance?	L3	CO2	**05**
		OR			
	b.	How do you define the voting mechanism in a majority vote classifier, and what considerations are important for implementing it?	L3	CO2	**05**
4.	**a.**	Demonstrate Q learning	L2	CO3	**05**
		OR			
	b.	How do high bias models typically behave, and what are the common indicators of underfitting?	L4	CO3	**05**
5.	**a.**	List out different performance evaluation metrics.	L2	CO4	**05**
		OR			
	b.	Difference between model validation and testing	L4	CO4	**05**
6.	**a.**	What are the key features of Flask that make it a popular choice for web application development?	L4	CO5	**05**
		OR			
	b.	What are the key steps involved in integrating a machine learning classifier into a web application?	L2	CO5	**05**
7.	**a.**	How do you evaluate the effectiveness of models trained using bootstrap aggregation compared to single models?	L5	CO2	**05**

		OR			
	b.	What considerations should be made for random seed selection in hold-out validation to ensure reproducibility of results?	L4	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	List the operations in computational graph. Give an example of creating a simple computational graph in TensorFlow.	L3	CO1	**10**
		OR			
	b.	Give an overview of working with batch and stochastic training, evaluating models	L2	CO1	**10**
9.	**a.**	How does a random forest differ from traditional bagging, and what unique features does it introduce?	L4	CO2	**10**
		OR			
	b.	Demonstrate AdaBoost ensemble with example	L3	CO2	**10**
10.	**a.**	Illustrate dueling network architecture, and how does it differ from traditional reinforcement learning architectures?	L3	CO3	**10**
		OR			
	b.	Give short notes on **i.** generalization, **ii.** feature selection, **iii.** modifying objective function	L2	CO3	**10**
11.	**a.**	How do you interpret a learning curve that shows high training accuracy but low validation accuracy? What does this indicate about the model?	L4	CO4	**10**
		OR			
	b.	Brief about ranking metrics, Classification metrics and regression metrics in detail	L2	CO4	**10**
12.	**a.**	Develop coding for insert, update, and delete records in a SQLite database using SQL commands?	L3	CO5	**10**
		OR			
	b.	Examine steps involved in transforming a regression model into a functional web application?	L5	CO5	**10**
13.	**a.**	Differences between AWS and Google Cloud in terms of services offered for application deployment?	L4	CO6	**10**
		OR			
	b.	Define backpropagation? How do you implement backpropagation from scratch, and what are the key components involved in the process? What are the common pitfalls when implementing backpropagation, and how can they be avoided?	L4	CO1	**10**